

DS	Session	Linked List	Question Information	Level 1	Challenge 31
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Question description

Professor Shiva decided to conduct an industrial visit for final year students, but he set a condition that if students received a passing grade in the surprise test, they would be eligible to go on the industrial visit.

He asked the students to study a topic linked list for 10 minutes before deciding to conduct a surprise test.

Professor-mandated questions, such as the deletion of nodes with a certain data D, are now being asked.

For example

if the given Linked List is 5->10->15->10->25 and delete after 10 then the Linked List becomes 5->15->25.

Constraints

1 < N < 100
1 < D < 1000

Input Format

First line contains the number of datas: N.

Second line contains N integers(the given linked list).

Next line indicates the node data D that has to be deleted.

Output Format

Single line represents the linked list after required elements deleted.

Linked List
Question No:1

Logical Test Cases

Test Case 1 INPUT (STDIN) 6 7 9 8 8 4 2 8 EXPECTED OUTPUT Linked List:->7->9->4->2	Test Case 2 INPUT (STDIN) 10 5 3 7 2 7 9 8 8 4 2 2 EXPECTED OUTPUT Linked List:->5->3->7->7->9->8->8->4
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Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  struct node
4  {
5      int data;
6      struct node *next;
7  };
8  struct node *createNode(int data)
9  {
10     struct node *newNode;
11     newNode = (struct node *)malloc(sizeof(struct node));
12     newNode->data = data;
13     newNode->next = NULL;
14     return newNode;
15 }
16 void insert(struct node **head, int data)
17 {
18     struct node *newNode;
19     struct node *temp;
20     newNode = createNode(data);
21     if (*head == NULL)
22     {
23         *head = newNode;
24         return;
25     }
26     temp = *head;
27     while (temp->next != NULL)
28     {
29         temp = temp->next;
```

```
30     }
31     temp->next = newNode;
32 }
33 /* Delete ALL occurrences of data */
34 void deleteAll(struct node **head, int data)
35 {
36     struct node *temp;
37     struct node *prev;
38     /* Delete matching nodes from the beginning */
39     while (*head != NULL && (*head)->data == data)
40     {
41         temp = *head;
42         *head = (*head)->next;
43         free(temp);
44     }
45     if (*head == NULL)
46         return;
47     prev = *head;
48     temp = (*head)->next;
49     while (temp != NULL)
50     {
51         if (temp->data == data)
52         {
53             prev->next = temp->next;
54             free(temp);
55             temp = prev->next;
```

```
56         }
57         else
58         {
59             prev = temp;
60             temp = temp->next;
61         }
62     }
63 }
64 void display(struct node *head)
65 {
66     struct node *temp;
67     temp = head;
68     printf("Linked List:");
69     while (temp != NULL)
70     {
71         printf("->%d", temp->data);
72         temp = temp->next;
73     }
74     printf("\n");
75 }
76 int main()
77 {
78     struct node *head = NULL;
79     int n;
```

```
80     int data;
81     int d;
82     int i;
83     scanf("%d", &n);
84     for (i = 0; i < n; i++)
85     {
86         scanf("%d", &data);
87         insert(&head, data);
88     }
89     scanf("%d", &d);
90     deleteAll(&head, d);
91     display(head);
92     return 0;
93 }
```

Output:

Test Case-1:

```
6
7 9 8 8 4 2
8
Linked List:->7->9->4->2
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
```

Test Case-2:

```
10
5 3 7 2 7 9 8 8 4 2
2
Linked List:->5->3->7->7->9->8->8->4
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
```

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